

NYC Bus Service Analysis for Hayden AI

Making NYC Move Faster

1. What Hayden AI Does :

Hayden AI mounts cameras on transit vehicles to automatically detect and enforce bus lane violations — clearing the way for buses to run on time.

2. The Business Problem

Cars illegally blocking bus lanes cause cascading delays. Without enforcement, buses miss scheduled trips, reducing rider trust and transit efficiency across the entire network.

3. My Solution

Using NYC's 10-year MTA dataset, I pinpointed the exact routes and boroughs where service delivery is lowest — giving Hayden AI a data-backed deployment priority list.

4. My Recommendations

- My analysis shows Queens had the highest missed trips at 678,000 over 10 years, with routes Q6 and BX15 showing the lowest service delivery in the entire dataset. I recommend Hayden AI prioritizes these routes for its first camera installations to show the fastest measurable improvement.
- Hayden AI should approach MTA with a weekday-only camera proposal. My data shows reliability drops consistently on weekdays across all boroughs while weekends perform better, same routes, same drivers, only the traffic changes. A weekday contract is cheaper for MTA to approve and directly targets when the problem actually happens.
- Local bus routes have the worst service delivery in my analysis, accounting for 78% of all missed trips. These routes run in mixed traffic with no lane protection unlike Select Bus which already has dedicated lanes. Hayden AI's cameras can fill that gap giving local routes the same protection without MTA having to spend money building new infrastructure.

HAYDEN AI - Helping NYC Move Faster

Jan 1, 2016 - Dec 31, 2025

NYC Bus Relia...
94.5%

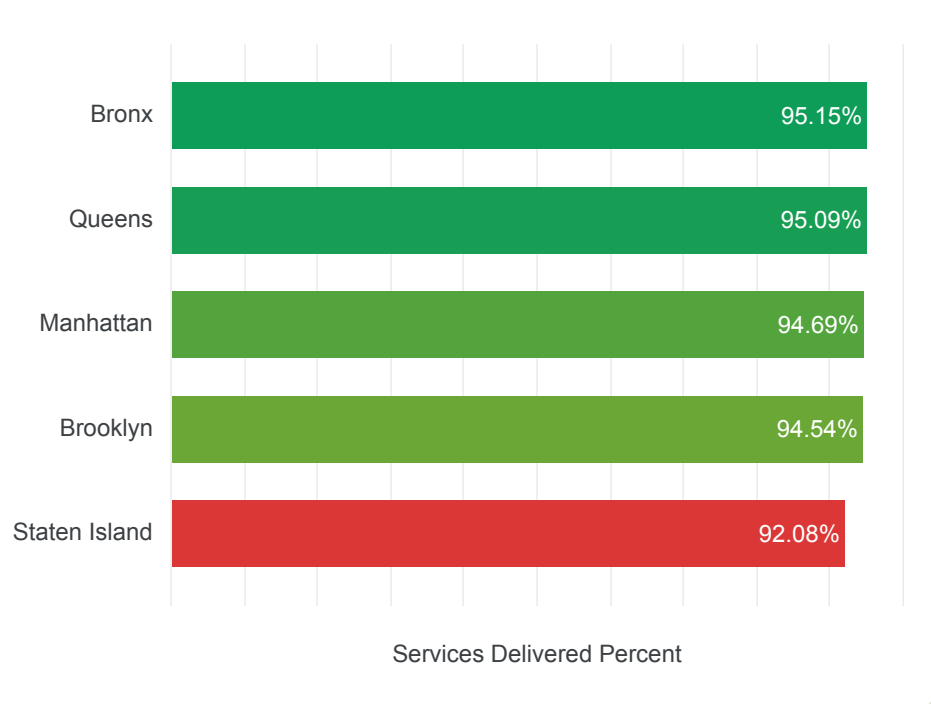
Peak Reliability
93.9%

Total Missed Trips
1,846,639

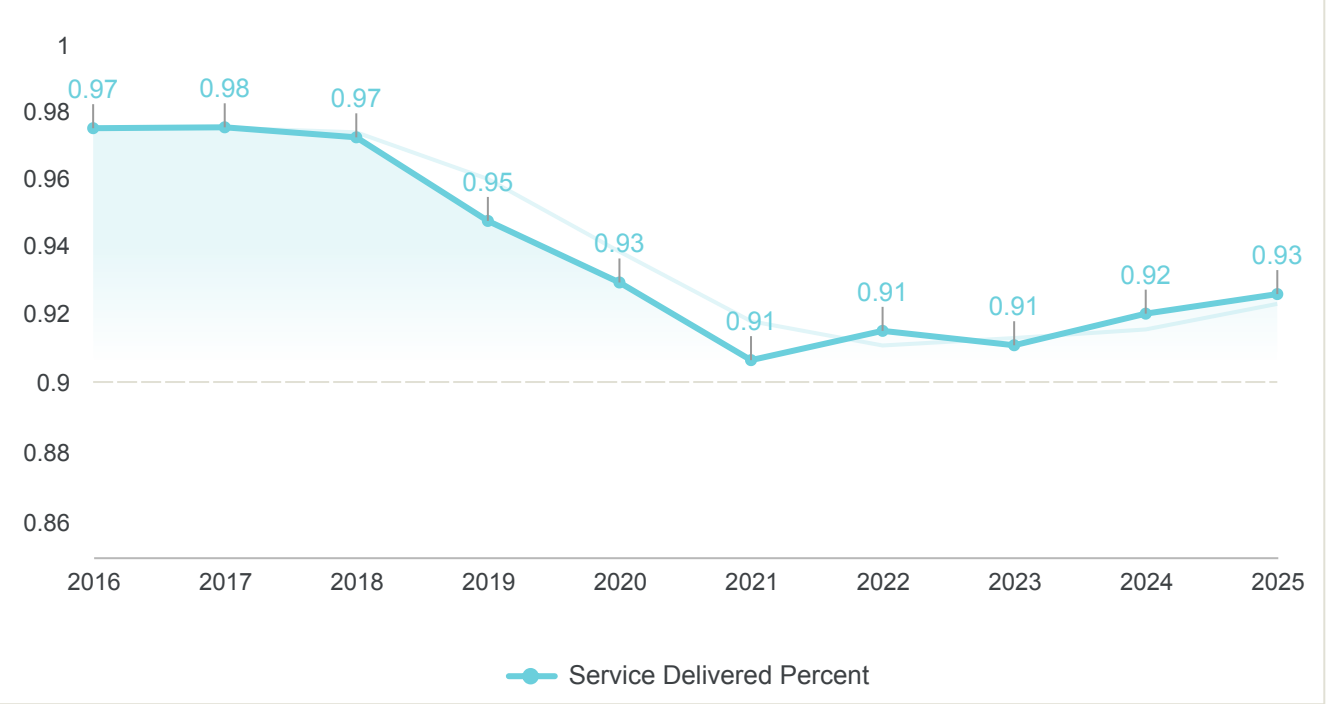
Lowest Avg Borough Reliability
Staten Island

Best Borough Reliability
Bronx

Borough Reliability Ranking



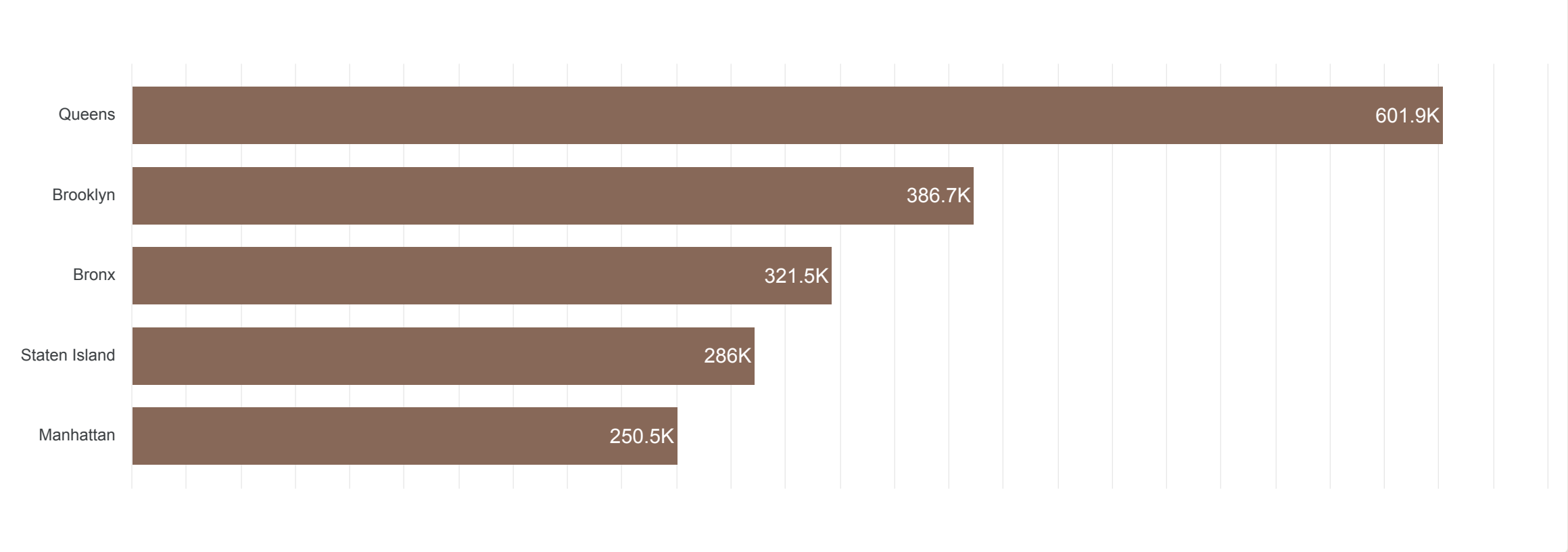
Bus Reliability Over Time (2015 -2026)



Staten Island is the least reliable borough at 92.4% — 2.8 points below Bronx. We need the enforcement here.

Service took a huge hit in 2020 and still hasn't fully bounced back 5 years later. This is not just a COVID problem it is more of a structural problem (Scope for Hayden AI)

Total Missed Bus Trips by Borough (2015 to 2026)



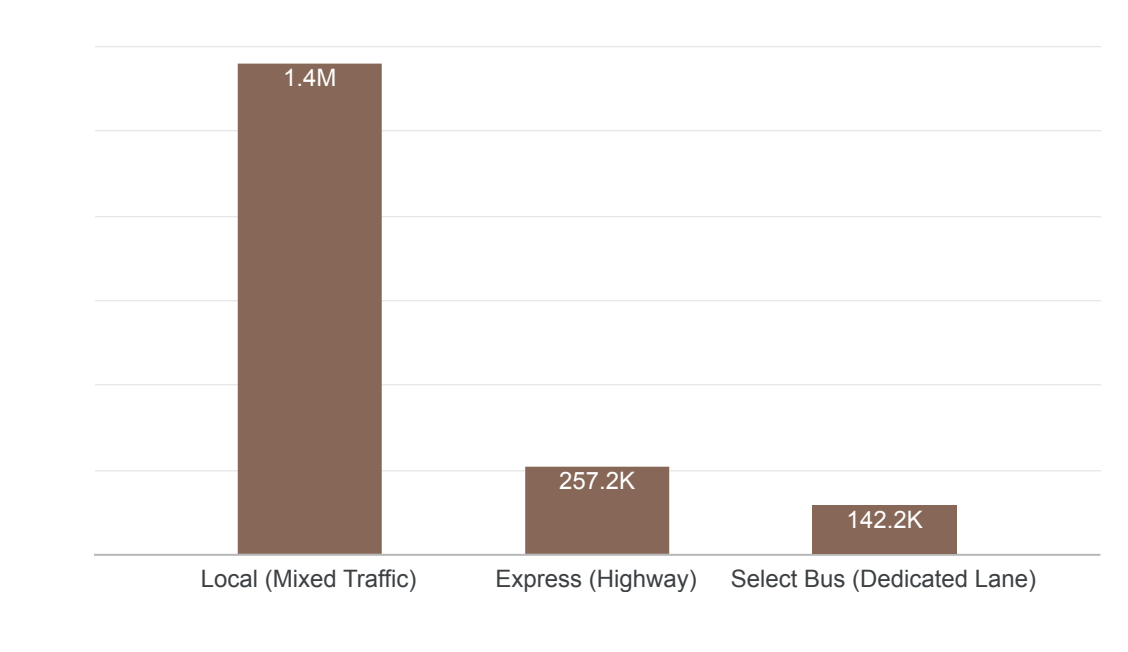
Queens has the highest missed trips at 678K — nearly double Manhattan. Despite Staten Island having the worst reliability rate, Queens has the largest enforcement impact by volume.

Highest Missed Trips by Route — Enforcement Priority List

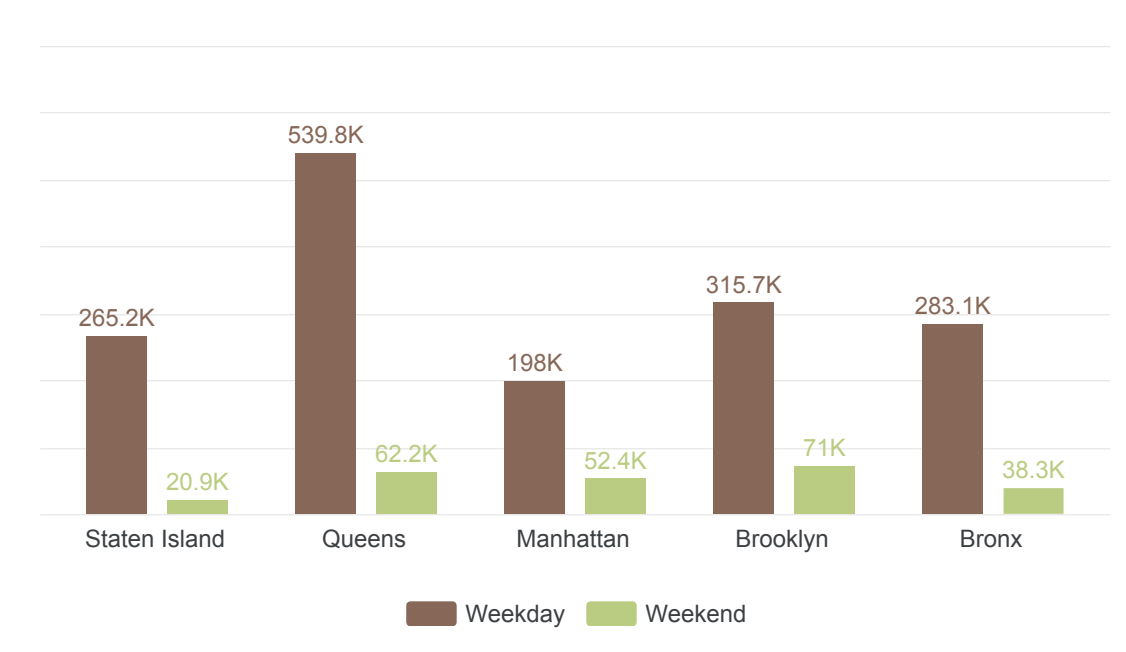
Route Name	Borough	Reliability	Scheduled Bus	Actual Buses	Missed Trips
Q6	Queens	91.8%	263,171	231,328	31843
BX15	Bronx	90.5%	228,918	199,939	28979
Q64	Queens	93.2%	250,515	227,024	23491
Q49	Queens	92.6%	223,466	200,774	22692
Q111	Queens	92.6%	231,083	211,601	19482
Q10	Queens	93.2%	212,508	193,285	19223
BX41+	Bronx	86.3%	126,880	108,345	18535
B103	Brooklyn	93.7%	242,236	224,342	17894
B46	Brooklyn	94.1%	214,652	199,216	15436
Q37	Queens	93.1%	153,290	138,177	15113
M101	Manhattan	89.6%	145,225	130,616	14609
B74	Brooklyn	86.3%	115,874	101,448	14426
B6	Brooklyn	95.4%	276,671	262,329	14342
B17	Brooklyn	92.2%	175,538	161,249	14289
B35	Brooklyn	93.2%	230,813	216,841	13972

These are the exact bus routes where Hayden AI cameras should go first. Q6 bus route alone had 34,749 buses that never showed up.

Total Missed Buses By Trip Type



Weekday vs Weekend Missed Trips



Local buses miss the most trips because they have zero lane protection. Select Bus routes — which have dedicated lanes — perform far better.

Buses are less reliable Monday to Friday in every single borough. Weekends have less traffic and better service — proving traffic is the problem